

CO3 - AT1

Case Study Analysis

Descriptive Statistics, Dispersion, Outliers & Distribution Shape

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Course: DSA0405 - Fundamentals of Data Sciences

Slot: D

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Methodology Note

Each case study below is solved using population-level descriptive statistics (mean, median, mode), measures of dispersion (range, variance, standard deviation, coefficient of variation) and quartile-based measures (Q1, Q3, IQR) where relevant. Outliers are flagged using the standard 1.5 x IQR rule: any value below $Q1 - 1.5(IQR)$ or above $Q3 + 1.5(IQR)$ is treated as an outlier. Variance and standard deviation are computed as population statistics (dividing by N) since each dataset represents the complete set of observations described in the scenario, not a sample drawn from a larger population.

$$\begin{aligned} \text{Mean } \mu &= \Sigma x / N & \text{Variance } \sigma^2 &= \Sigma (x - \mu)^2 / N & \text{Std. Dev. } \sigma &= \sqrt{\sigma^2} \\ \text{IQR} &= Q3 - Q1 & \text{Outlier bounds} &= [Q1 - 1.5 \times IQR, Q3 + 1.5 \times IQR] & z &= (x - \mu) / \sigma \end{aligned}$$

Case Study 1 — Employee Salaries

Data (₹ lakhs, n = 9): 4, 5, 5, 6, 7, 8, 8, 9, 30

Mean	Median	Mode	Range	Variance	Std. Dev.
9.11	7	5	26	56.99	7.55

Outlier

The value 30 is a clear outlier — more than three times the next-highest salary and over 22 lakhs above the median. It lies far outside the IQR-based bound of 12.5 lakhs ($Q1 = 5$, $Q3 = 8$, $IQR = 3$).

Best Measure of Central Tendency

The median (₹7 lakhs) best represents the typical salary. The mean (₹9.11 lakhs) is pulled upward by the single outlier and overstates what a typical employee earns — in fact 8 of the 9 employees earn below the mean. The mode (₹5 lakhs, appearing twice) is too low and unstable to be a reliable summary. Because the median is resistant to extreme values, it gives the most honest picture of central salary in a right-skewed, outlier-affected dataset like this one.

Case Study 2 — Website Response Time

Data (ms, n = 10): 120, 125, 128, 130, 132, 135, 140, 145, 150, 300

Mean	Median	Range	Variance	Std. Dev.
150.5 ms	133.5 ms	180 ms	2558.05	50.58 ms

Skewness

The data is right-skewed (positively skewed). Nine of the ten requests cluster tightly between 120 and 150 ms, while one extreme value (300 ms) stretches the right tail. This pulls the mean (150.5 ms) well above the median (133.5 ms), the classic signature of right skew.

Why the Median May Be More Useful

The median is unaffected by the single slow request and reflects the response time a typical user actually experiences (≈134 ms). The mean is distorted by the 300 ms outlier and would lead an engineer to believe the "typical" response is far slower than it really is — a misleading basis for setting performance targets or SLAs.

Case Study 3 — Student Performance

Data (marks, n = 12): 42, 45, 48, 50, 52, 52, 55, 58, 60, 62, 65, 95

Q1	Q2 (Median)	Q3	IQR	Mean
49.5	53.5	60.5	11	57.0

Outlier Check (1.5 × IQR Rule)

Lower bound = $Q1 - 1.5(IQR) = 49.5 - 16.5 = 33.0$. Upper bound = $Q3 + 1.5(IQR) = 60.5 + 16.5 = 77.0$. The score of 95 lies above the upper bound and is flagged as an outlier; no value lies below the lower bound.

Effect of the Outlier on the Mean

The mean (57.0) sits noticeably above the median (53.5) because the outlier score of 95 pulls the arithmetic average upward, even though it represents only one of twelve students. Since the mean uses every value with equal weight, a single extreme

score has an outsized effect on it, while the median — based only on rank position — stays close to where most students actually scored.

Case Study 4 — E-Commerce Orders

Data (orders/day, $n = 10$): 18, 20, 22, 22, 24, 25, 26, 28, 30, 35

Mean	Median	Mode	Variance	Std. Dev.	CV
25.0	24.5	22	22.80	4.78	19.1%

What the Standard Deviation Tells the Company

A standard deviation of about 4.78 orders means daily order counts typically fall within roughly ± 5 orders of the 25-order average — i.e. most days see between about 20 and 30 orders. It quantifies day-to-day unpredictability in demand, which matters for staffing and inventory planning.

Variability Assessment

The coefficient of variation, $CV = (\sigma / \mu) \times 100 = (4.78 / 25.0) \times 100 \approx 19.1\%$, falls in the moderate range (commonly, CV under $\sim 15\%$ is considered low variability and above $\sim 30\%$ is considered high). At 19.1%, the order data is best described as moderately variable — demand fluctuates from day to day but without wild, unpredictable swings.

Case Study 5 — Comparing Two Data Science Models

Model A errors ($n = 8$): 2, 3, 3, 4, 4, 5, 5, 6

Model B errors ($n = 8$): 1, 1, 2, 4, 5, 7, 8, 9

Model	Mean Error	Variance	Std. Dev.
Model A	4.000	1.500	1.225
Model B	4.625	8.734	2.955

Consistency

Model A has both the lower variance (1.50 vs 8.73) and lower standard deviation (1.23 vs 2.96), so its prediction errors are tightly clustered around its mean. Model A is the more consistent, more predictable model.

Does a Lower Mean Error Mean "Better"?

Not necessarily. Model A also happens to have the lower mean error here, but mean error and consistency are two different properties: a model could have a low mean error purely by having a few very small errors offset by a few large ones, while a slightly higher-mean model with low variance behaves reliably every time. In applications where predictable, stable performance matters (e.g. safety-critical systems), a model with marginally higher mean error but much lower variance can be the better practical choice.

Case Study 6 — Detecting an Unusual Customer

Website-wide: mean spend $\mu = ₹2,000$, standard deviation $\sigma = ₹400$. Customer spend $x = ₹3,200$.

$$z = (x - \mu) / \sigma = (3200 - 2000) / 400 = 3.0$$

Interpretation

- The customer's z-score is 3.0 — their spending is 3 standard deviations above the average.
- Yes, this is unusually high. A z-score beyond ± 2 (and especially ± 3) is statistically rare under a roughly normal distribution — fewer than about 0.3% of customers would be expected to spend this much or more.
- A negative z-score would mean the customer spent below the site-wide average — e.g. $z = -1$ would mean spending one standard deviation (₹400) below ₹2,000, i.e. ₹1,600.
- A customer with $z = -1.5$ spent 1.5 standard deviations below average: $₹2,000 - 1.5(₹400) = ₹1,400$ — a notably low-spending, though not extreme, customer.

Case Study 7 — Delivery Time Analysis

Data (minutes, n = 10): 25, 27, 28, 29, 30, 30, 31, 32, 34, 60

Mean	Median	Q1	Q3	IQR
32.6 min	30.0 min	28.25 min	31.75 min	3.5 min

Outlier Check

Bounds = $[Q1 - 1.5(IQR), Q3 + 1.5(IQR)] = [23.0, 37.0]$. The delivery of 60 minutes falls well above the upper bound and is a clear outlier — likely a delayed or exceptional delivery rather than typical performance.

Mean vs. Median

The median (30 minutes) gives a far better description of a typical delivery. The mean (32.6 minutes) is inflated by the single 60-minute outlier, making deliveries look slower on average than what 9 of the 10 customers actually experienced. For operational reporting (e.g. "expected delivery time"), the median avoids being skewed by rare delays and better reflects customer experience.

Case Study 8 — Two Cities' Temperatures

City A (°C, n = 7): 28, 29, 30, 30, 31, 30, 32

City B (°C, n = 7): 20, 25, 30, 35, 30, 25, 45

City	Mean	Variance	Std. Dev.
City A	30.0 °C	1.43	1.20 °C
City B	30.0 °C	57.14	7.56 °C

Greater Variability

City B has far greater temperature variability — its variance (57.14) is roughly 40 times that of City A (1.43), driven by swings as large as 45°C.

Same Mean, Different Std. Dev. — Why?

The mean only describes the central location of a dataset; it says nothing about how spread out the individual values are around that centre. Two cities can average the same 30°C while one hovers consistently near that value (City A) and the other swings widely from cold to hot days (City B). Standard deviation captures exactly this dispersion that the mean cannot.

More Predictable City

City A is far more predictable — its low standard deviation (1.20°C) means daily temperatures stay close to 30°C, whereas City B's high standard deviation (7.56°C) makes its day-to-day temperature much harder to forecast.

Case Study 9 — Data Center CPU Usage

Data (% usage, n = 10): 40, 42, 45, 45, 46, 48, 50, 52, 55, 90

Mean	Median	Mode	Range	Q1	Q3	IQR
51.3%	47.0%	45%	50	45.0	51.5	6.5

Outliers

Bounds = $[45.0 - 1.5(6.5), 51.5 + 1.5(6.5)] = [35.25, 61.25]$. The 90% reading lies well above the upper bound and is a clear outlier — a CPU usage spike well beyond normal operating levels.

Skewness

The distribution is right-skewed (positively skewed): the mean (51.3%) sits above the median (47.0%) because the extreme 90% spike drags the average upward while most readings cluster in the 40–55% range. This pattern — a long right tail caused by occasional load spikes — is typical of CPU/utilization monitoring data.

Case Study 10 — Comparing Two Datasets

Dataset A (n = 5): 10, 20, 30, 40, 50

Dataset B (n = 5): 28, 29, 30, 31, 32

Dataset	Mean	Variance	Std. Dev.
A	30.0	200.0	14.14
B	30.0	2.0	1.41

Spread and Consistency

Both datasets share an identical mean of 30, but Dataset A has dramatically greater spread — its variance (200.0) is 100 times that of Dataset B (2.0). If these represent model errors, Dataset B's model is far more consistent: its errors stay tightly within ± 1.4 of the mean, whereas Dataset A's errors range from 10 to 50.

Why Mean Alone Is Not Enough

The mean alone would suggest these two datasets are identical, since both average exactly 30. But it reveals nothing about reliability or risk: Dataset A's values are wildly dispersed while Dataset B's are tightly clustered. Any decision based only on the mean — comparing model quality, judging consistency, or assessing risk — would be dangerously incomplete without also considering variance/standard deviation to describe how much the data spreads around that mean.